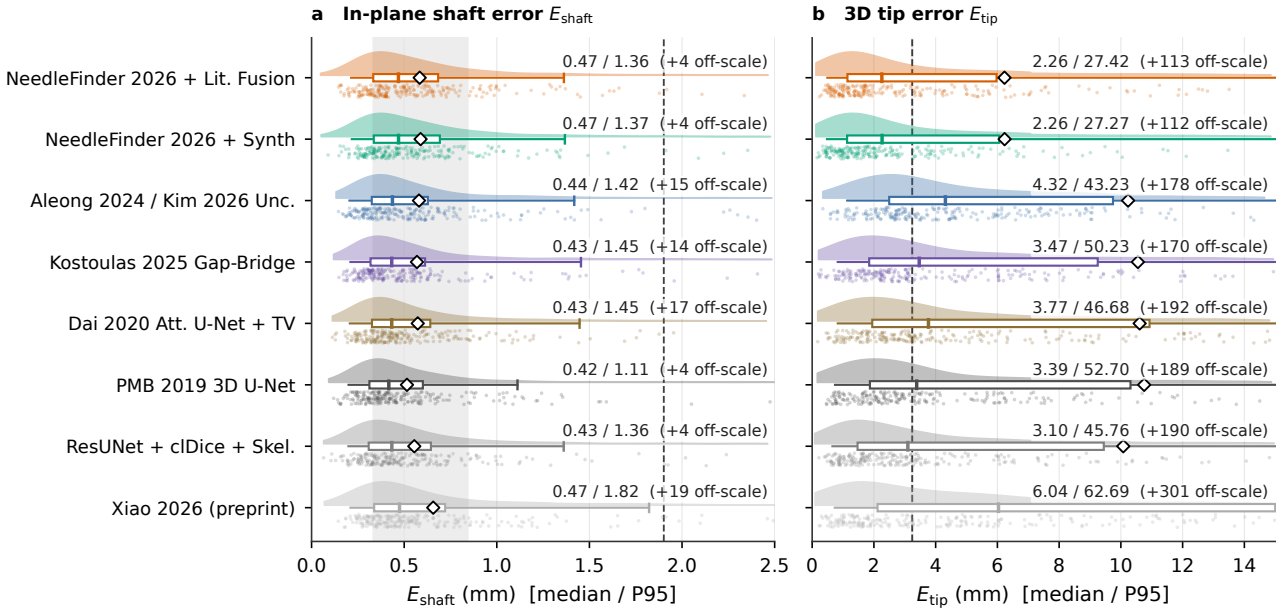

a In-plane shaft error E_{shaft}
b 3D tip error E_{tip}
NeedleFinder 2026 + Lit. Fusion
NeedleFinder 2026 + Synth
Aleong 2024 / Kim 2026 Unc.
Kostoulas 2025 Gap-Bridge
Dai 2020 Att. U-Net + TV
PMB 2019 3D U-Net
ResUNet + clDice + Skel.
Xiao 2026 (preprint)
0.47 / 1.36 (+4 off-scale)
0.47 / 1.37 (+4 off-scale)
0.44 / 1.42 (+15 off-scale)
0.43 / 1.45 (+14 off-scale)
0.43 / 1.45 (+17 off-scale)
0.42 / 1.11 (+4 off-scale)
0.43 / 1.36 (+4 off-scale)
0.47 / 1.82 (+19 off-scale)
2.26 / 27.42 (+113 off-scale)
2.26 / 27.27 (+112 off-scale)
4.32 / 43.23 (+178 off-scale)
3.47 / 50.23 (+170 off-scale)
3.77 / 46.68 (+192 off-scale)
3.39 / 52.70 (+189 off-scale)
3.10 / 45.76 (+190 off-scale)
6.04 / 62.69 (+301 off-scale)
0.0 0.5 1.0 1.5 2.0 2.5
E_{shaft} (mm) [median / P95]
0 2 4 6 8 10 12 14
E_{tip} (mm) [median / P95]